

L3 capability selection — EDA

Question: for the selected stages of an approved value stream, does the model pick the right **L3 business capabilities** — and where does it fail?

Task — classification, one level below stages

L3 selection mirrors stage selection one level deeper: for **each selected stage**, choose which of that stage's **governed L3 capabilities** the idea exercises. The model picks only from that stage's own candidate list (never invents, renames, or borrows across stages). It's the **finest-grained** classification in the pipeline (L3 within stages within value streams).

Setup

- **GT** = the Theme's `L3 Business Capability Model` field (`CAP#####` ids), recorded at the **theme level** (not per stage).
- Generation is **per stage** (raw idea card only, no signals — the locked theme-gen decision), so we **aggregate** the generated L3 across the value stream's GT stages into a theme-level set and score it against the GT theme set.
- **Two modes**, like stages: `per_stage` (one call per stage) vs `one_call` (batched — production).
- Sample: 25 tickets, seed 13, ~231 scored value streams.

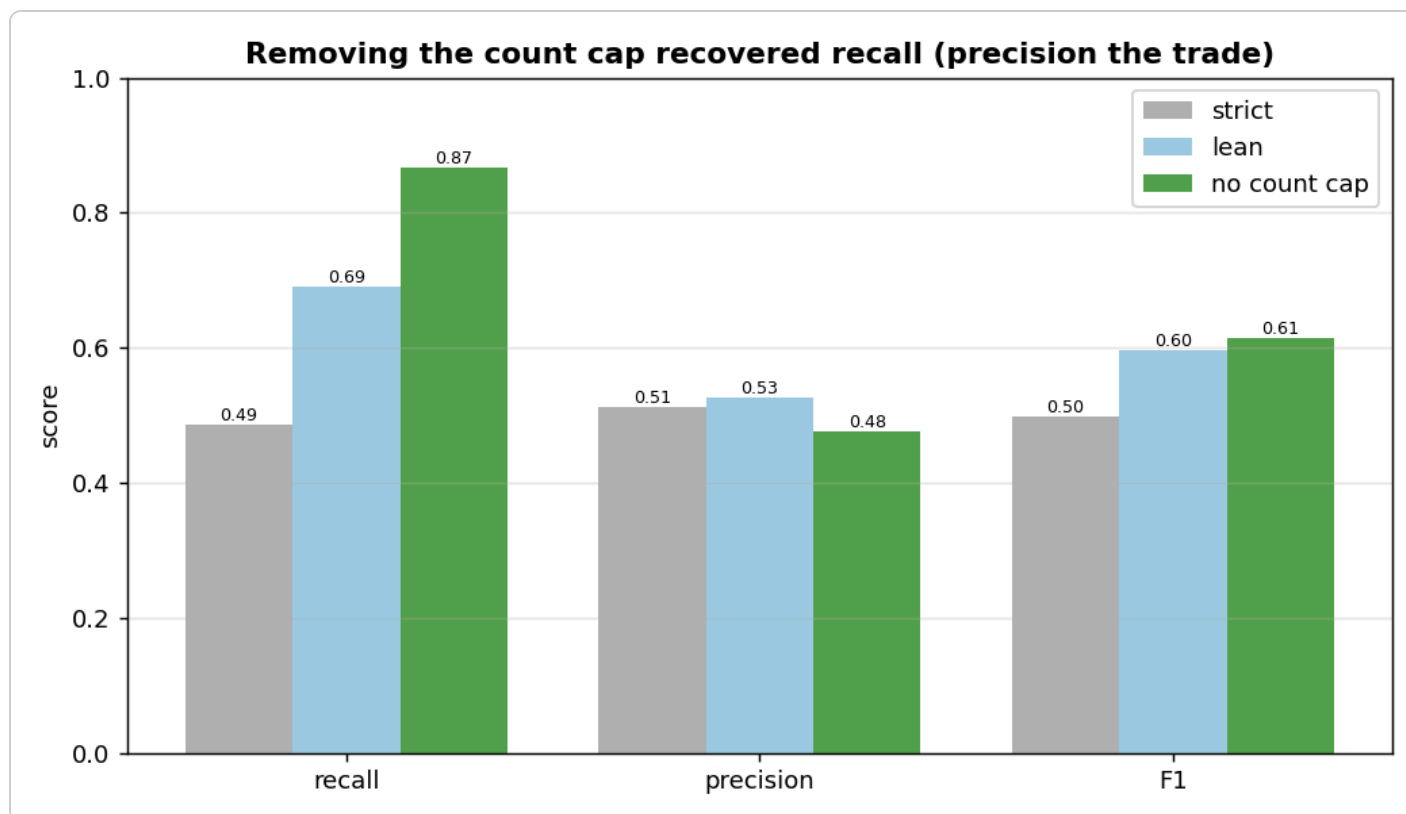
The coverage ceiling — theme-vs-stage granularity (read first)

	value
GT L3 reachable from the GT stages	58%

GT L3 is tagged at the **theme** level, but the catalogue maps each L3 to a **specific stage**. So **42% of theme GT L3 belong to stages outside the selection** and can't be predicted from the GT stages' candidate lists. We therefore **score only the answerable subset** (`theme GT L3 n the GT stages' candidates`) — the real model recall — and report this 58% separately as the **ceiling**. It is a GT/granularity mismatch, not a model failure.

Finding 1 — the count cap was suppressing recall (the big lever)

The first run was poor: **one_call F1 0.50, recall 0.49**. The drop diagnosis showed **49% of dropped answerable GT L3 were** `context_present_but_dropped` — the card supports them, the model saw them, and dropped them anyway. Cause: the prompt's soft cap ("*typically 1-4 per stage*"). When a stage's GT has more than that, the model stays "in range" and drops card-supported capabilities — the same *count-is-the-lever* pattern found in value-stream selection.

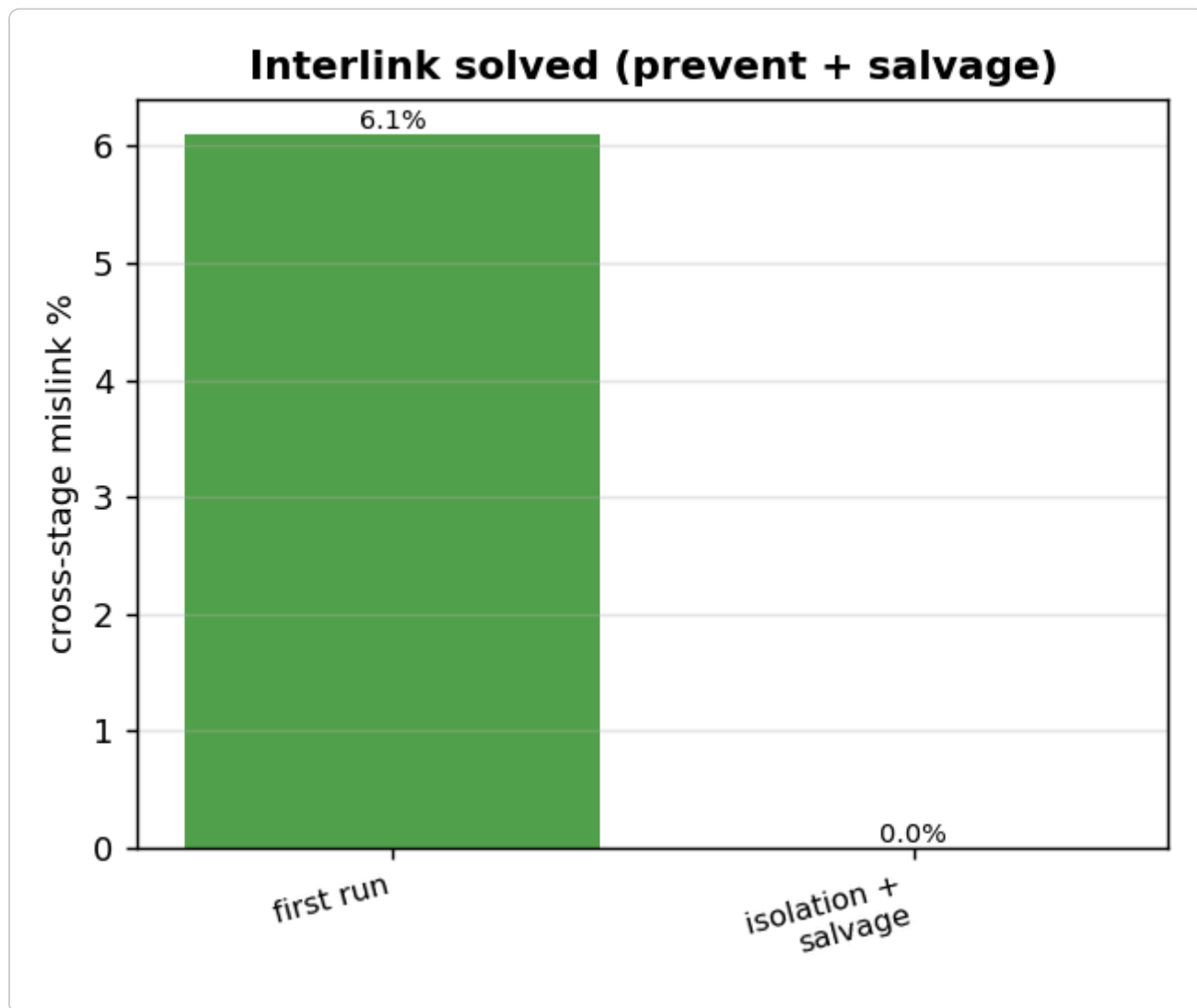


one_call across the prompt versions:

version	precision	recall	F1
strict (default EXCLUDE, require exact phrase)	0.513	0.486	0.499
lean (include what the work exercises)	0.526	0.691	0.597
no count cap (include EVERY capability, no quota)	0.476	0.867	0.615

Removing the cap lifted **recall 0.49 → 0.87** — a large recovery, with precision trading down to ~0.48 (the cap had been suppressing some false positives too). F1 net **+0.12**. The `context_present_but_dropped` bucket **halved (49% → 28%)**, confirming the cap was the cause.

Finding 2 — interlink solved: prevent + salvage → 0

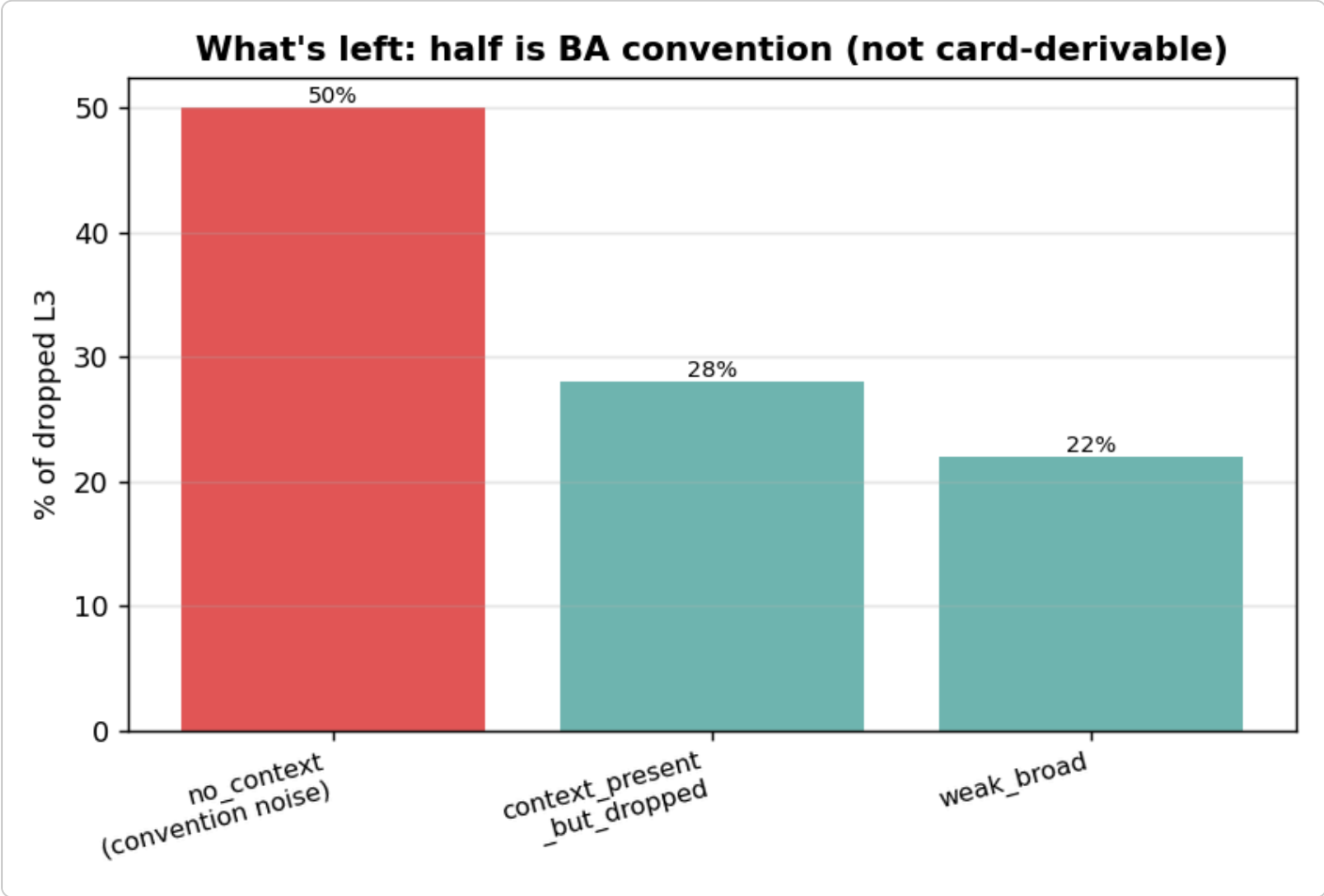


The batched call can put an L3 under the wrong stage (cross-stage mislink) — 6.1% at first. Two layers fixed it:

- **Prevent (prompt):** a strict-isolation block — *check the exact id is printed under THAT stage's list; similar-named capabilities exist under several stages, use only the printed id.*
- **Salvage (code):** a capability id is governed by exactly one stage, so a mislinked pick is **reassigned to its true owner stage** instead of dropped.

Result: **cross-stage mislink 6.1% → 0%.** The prompt prevented it; the salvage net recovers any that ever slip through (and recovers the mislinked-but-valid L3 into the prediction rather than losing it).

Finding 3 — what's left is mostly BA convention, not a model gap



After the recall fix, the remaining drops (127 answerable GT L3) split as:

grounding	share	meaning
no_context_for_capability	50%	the card has no evidence — the BA tagged it by capability-model convention (label noise, not derivable)
context_present_but_dropped	28%	card supports it, dropped anyway — the residual fixable bit
weak_broad_context	22%	borderline

So **half the remaining drops are un-derivable from the ticket** (BA convention), and the fixable bit shrank to 28%. Pushing recall further would keep trading precision for diminishing, convention-capped gains — recall is **topped at the right place**.

Verdict — locked

L3 capability selection (answerable): F1 $\approx$ 0.62, recall $\approx$ 0.87, precision $\approx$ 0.48, on a 58% coverage ceiling (theme-vs-stage granularity), 0% cross-stage mislink.

- It's the **finest-grained** classification in the pipeline, against **theme-level GT** that doesn't decompose to stages — so the 58% coverage is a structural ceiling, and we score the answerable L3.

- **Removing the count cap was the lever** — recall $0.49 \rightarrow 0.87$ (the *count-is-the-lever* pattern again); precision is the accepted trade.
- **Interlink solved** — strict-isolation prompt (prevent) + salvage (correct) $\rightarrow$ 0% mislink.
- The residual gap is **mostly BA capability-model convention** (50% no-context) the card can't derive, plus a small fixable tail.

No further changes.